King Fahd University of Petroleum and Minerals
College of Computing & Mathematics
Information & Computer Science Department

Decentralized Document Verification on Bitcoin Using Taproot

Master Thesis Proposal

Submitted by:

Juan Crespo

Email: g202393610@kfupm.edu.sa

Supervised By:

Dr. Sultan Al-Muhammadi

December 2025

Abstract

Document verification is a fundamental mechanism for detecting document falsification and fraud in institutional processes. Traditional approaches are based on centralized infrastructures, which incur high operational costs, limited availability, and single points of failure. When the issuing organization is compromised or becomes unavailable, verification capability is effectively lost.

Bitcoin, introduced in 2009, operates as a decentralized network secured by a proof-of-work consensus protocol. Its properties of immutability, high availability, and permissionless access make it a suitable foundation for applications beyond monetary use.

Bitcoin has been widely used for anchoring non-repudiable information. Several data anchoring techniques have been proposed for Bitcoin, including OP_RETURN, fake addresses, OpenTimestamps, and Script- and Witness-based methods. However, these approaches exhibit limitations in scalability, privacy, and usability. The activation of the Taproot upgrade in 2021 introduced more efficient and private cryptographic primitives, enabling new application designs at the base layer.

This work proposes a novel protocol for decentralized document verification using Taproot. The protocol supports integrity verification, issuer authenticity verification, and revocation status checking. By anchoring a batch of N documents to a single pay-to-taproot P2TR output, the system improves scalability and preserves privacy. Verification is performed entirely off-chain, and the protocol evaluates both computational costs and on-chain block space usage.

Table of Contents

1 Introduction **1**

2 Literature Review **1**

2.1 Data Anchoring Techniques 2

2.2 Cryptographic Foundations 3

2.3 Alternative Systems and Applications 4

3 Motivation **5**

4 Problem Statement **6**

4.1 Research Questions 6

4.2 Specific Objectives 6

5 Methodology **7**

5.1 Phase I: Protocol Design and Architecture 7

5.1.1 On-chain vs Off-chain Components 7

5.1.2 Integrity Verification 9

Leaf script design (to be analyzed). 9

Validation proof contents (integrity). 9

Integrity check (verifier). 10

5.1.3 Key Separation 10

5.1.4 Issuer Authenticity Verification (Signature) 10

Signature generation (issuer). 11

Authenticity check (verifier). 11

5.1.5 Document Revocation 11

5.2 Phase II: Proof of Concept (PoC) 12

5.3 Phase III: Performance Estimation and Analysis 13

6 Timeline **13**

6.1 Gantt Chart Representation 14

7 Conclusion **14**

References **15**

1 Introduction

Documents issued by an organization serve as proof of a factual occurrence. Verifying the authenticity, integrity, and non-revocation of such documents relies on the issuer. Trust in the organization depends on the efficiency and accessibility of this procedure.

Historically, document verification was implemented through centralized physical processes. These systems require significant operational costs to achieve availability and efficiency. With the advent of the Internet and increased computational capacity, issuance and verification transitioned to centralized digital systems, introducing additional operational costs related to server infrastructure, cybersecurity, and software licensing.

Bitcoin launched in 2009 as the first decentralized network. It has experienced no censorship events or downtime as of this writing. It supports limited smart contract execution under the unspent transaction output UTXO model, enabling information anchoring. Constrained by design to mitigate denial-of-service DoS vectors and high block space costs, limiting the range of application logic that can be implemented directly on-chain.

Several studies have explored alternative blockchain platforms, such as Ethereum, to enable more expressive application logic. Their use of Turing-complete smart contracts allows on-chain certificate validation and verification. However, this approach introduces additional trust assumptions and higher consensus complexity, increasing exposure to censorship and state reorganization risks, reducing verification autonomy.

The Bitcoin protocol has evolved over 16 years through major community-accepted upgrades. SegWit [1] and Taproot [2] improved security, privacy, and block space efficiency, expanding the set of feasible base-layer applications, especially those requiring efficient data commitments and verification.

2 Literature Review

A Bitcoin transaction is a variable-sized data structure with a theoretical maximum size of 1 MB. It consists of multiple fields, including version information, inputs, outputs, and locking scripts. The `scriptPubKey` field in transaction outputs defines the spending conditions and determines the transaction type.

The transaction format also enables data embedding and cryptographic commitments

through specific output constructions. This capability has been widely used as a basis for information anchoring, enabling verifiable integrity guarantees without modifying the underlying protocol.

2.1 Data Anchoring Techniques

Although Bitcoin was originally designed for peer-to-peer monetary transfers, its globally replicated and immutable ledger motivated its use for applications beyond payments. Lemieux [3] evaluates blockchain technology as a mechanism for preserving the integrity of digital records under various ISO standards. The study concludes that while blockchain-based systems can provide short- and medium-term integrity guarantees, they do not ensure initial record reliability and present limitations as long-term preservation mechanisms. In particular, the analysis highlights the long-term reliance on cryptographic primitives such as hash functions and digital signatures, including SHA-256 and RIPEMD-160, as a fundamental concern.

Except for coinbase transactions, Bitcoin does not natively provide a designated field for arbitrary metadata. Gao and Nobuhara [4] propose embedding information into standard P2PKH transactions by encoding metadata within so-called fake addresses. By creating a transaction with N outputs, this approach allows the insertion of up to $20 \times N$ bytes. However, the method permanently destroys the associated Bitcoin by rendering the outputs unspendable, resulting in high economic cost and limited scalability.

Sward et al. [5] conduct a comprehensive analysis of data insertion techniques in Bitcoin transactions. Their study categorizes approaches into ScriptHash-based constructions and fake address techniques. While ScriptHash-based approaches are not constrained by a fixed data size limit, they require non-standard scripts and increased construction complexity. The authors conclude that no universally optimal data insertion method exists, as all approaches involve trade-offs between data capacity, transaction cost, network health, and standardness.

Bartoletti et al. [6] further analyze metadata embedding techniques and their impact on Bitcoin's primary function. They identify Unspent Transaction Output (UTXO) set growth as a critical concern, as excessive UTXO expansion negatively affects node performance. Among the evaluated methods, approaches based on `OP_RETURN` minimize UTXO bloat by explicitly marking outputs as provably unspendable.

The `OP_RETURN` opcode enables arbitrary data insertion by terminating script execution. Strehle and Steinmetz [7] analyze over 32 million transactions and demonstrate that `OP_RETURN` has become the dominant mechanism for embedding non-monetary data in Bitcoin. The study shows that most such transactions originate from a small number of services, notably Omni and Veriblock. Widely deployed protocols such as Blockcerts [8] and OpenTimestamps [9] rely on `OP_RETURN`-based anchoring. Despite its efficiency and standardization, `OP_RETURN` explicitly reveals the presence and location of embedded metadata, limiting privacy for sensitive verification use cases.

Legacy data anchoring techniques therefore exhibit two primary limitations: lack of privacy and transaction malleability. While SegWit effectively resolves malleability, legacy approaches still expose anchored data at the protocol level. Kedziora et al. [10] analyze the impact of SegWit adoption, showing that segregating witness data improves both transaction efficiency and security. They report transaction cost reductions of up to 75% and expanded data capacity through the Witness structure without increasing UTXO bloat.

Table 1 summarizes the main characteristics of Bitcoin-based data anchoring techniques discussed in the literature. The values are approximate and derived from the referenced works, providing a qualitative comparison of trade-offs.

Characteristic	**Fake Address**	**OP_RETURN**	**OpenTimestamps**	**Script Methods**	**Witness Methods**
Size (Bytes)	$20 \times N$	~80	~40	~1000	~4000
Privacy	No	Hash-visible	Hash	Hidden until spend	Hidden until spend
Scalability	Limited	Merkle Tree	Merkle Tree	Complex	Complex
UTXO Impact	High bloat	None	None	Low	Low
Cost Model	Fee + Burn	Standard	Aggregated	Low	Discounted

Table 1: Comparison of Bitcoin-based data anchoring techniques

2.2 Cryptographic Foundations

Advances in cryptographic primitives further contribute to the evolution of blockchain-based verification systems. Gayoso et al. [11] and Fang et al. [12] survey cryptographic foundations used in blockchain systems, emphasizing the role of hash functions, Merkle trees, and digital signatures. Yang et al. [13] analyze advanced signature constructions, highlighting efficiency and verifiability properties relevant to non-repudiation systems.

More recent work by Zhang et al. [14] demonstrates the efficiency and aggregation advantages of Schnorr-based signatures. These properties motivate their adoption in

Bitcoin through BIP340 [15] and their integration into Taproot.

Building upon these developments, BIP341 introduces Taproot as a soft fork extension of SegWit. Taproot defines a new output type, Pay-to-Taproot (P2TR), combining Schnorr signatures with Merkleized Abstract Syntax Trees (MAST). Since its activation in November 2021, Taproot has been successfully deployed on the Bitcoin network, enabling more private and efficient spending conditions. Despite its expressive power, the literature has not yet explored Taproot as the foundation for a complete document verification protocol encompassing integrity, authenticity, and revocation.

2.3 Alternative Systems and Applications

To overcome Bitcoin's script limitations, alternative blockchain platforms have adopted different execution models. Ethereum replaces the UTXO paradigm with an account-based model and supports a Turing-complete virtual machine.

Numerous studies leverage Ethereum smart contracts for certification and attestation. Works by Wang et al. [16], Li et al. [17], Pennino et al. [18], Palmisano et al. [19], Mishra et al. [20], and Harer et al. [21] demonstrate the flexibility of smart contracts for managing identity, permissions, and document validation. However, these systems incur increased execution cost, gas fees, and implementation complexity.

Beyond academic credentials, blockchain-based certification has been explored in industrial and supply-chain contexts. Hawashin et al. [22] analyze NFT-based certification for UAV manufacturing components, highlighting scalability and traceability requirements beyond education-focused use cases.

The institutional use of blockchain-based document verification has also been studied extensively. Castro and Au-Yong-Oliveira [23] analyze blockchain adoption for higher education diplomas, identifying economic incentives and adoption barriers. Rustemi et al. [24] provide a systematic review showing that most academic certificate verification platforms are implemented on Ethereum. They identify revocation as a major challenge, as modifying immutable smart contracts or credential hashes directly impacts identity consistency.

Recent studies further highlight unresolved challenges related to privacy and revocation. Ramica et al. [25] survey selective disclosure mechanisms in digital credentials, while Zhang et al. [26] analyze privacy-preserving revocation schemes in self-sovereign identity

systems. These works demonstrate that privacy-aware revocation remains a complex problem, often requiring additional infrastructure or off-chain coordination.

In summary, the literature reveals a fundamental trade-off. Bitcoin-based approaches relying on legacy mechanisms such as `OP_RETURN` provide low-cost anchoring but suffer from limited privacy and scalability. Smart contract platforms such as Ethereum enable expressive verification logic but incur higher execution costs and increased complexity. Recent advances introduced by Taproot suggest that this trade-off may be mitigated at the Bitcoin base layer, motivating further investigation into privacy-preserving and scalable document verification protocols.

3 Motivation

The primary motivation for this work arises from the fragility of centralized document verification models. When an issuing organization becomes unavailable, ceases operations, or loses its internal records, the ability to verify previously issued documents is effectively lost. This dependency introduces a single point of failure that undermines long-term verifiability and trust.

In contrast, the reliability of a decentralized system is rooted in its consensus protocol. Bitcoin operates under a Proof-of-Work (PoW) consensus mechanism, providing strong security guarantees, immutability, and resistance to denial-of-service (DoS) attacks. These properties make Bitcoin particularly suitable for applications that require long-term availability and independent verification.

Other blockchain-based solutions for document verification often rely on alternative networks such as Ethereum. While these platforms benefit from Turing-complete smart contracts and flexible execution models, they also introduce different scalability, cost, and trust trade-offs compared to Bitcoin. In many cases, verification logic is executed on-chain, increasing operational complexity and dependency on the continued availability of smart contract infrastructure.

The activation of Taproot significantly improves Bitcoin's privacy and block space efficiency at the base layer. These advancements enable new application designs that were previously impractical on Bitcoin. Motivated by these developments, this work explores a document verification protocol that operates entirely on Bitcoin's main layer, without

requiring additional protocol upgrades, second-layer solutions, or reliance on external blockchain networks.

4 Problem Statement

Existing decentralized approaches to document verification exhibit limitations that depend strongly on the underlying blockchain infrastructure. In the context of Bitcoin, current solutions based on protocols such as OpenTimestamps, Blockcerts, and `OP_RETURN` either expose embedded metadata, compromising privacy, or incur high costs in terms of block space and transaction fees. As a result, these approaches do not scale efficiently for large batches of documents or long-term institutional use.

The Taproot upgrade introduces new cryptographic primitives that significantly improve efficiency and privacy at the Bitcoin base layer. By adopting Schnorr signatures, Taproot enables more compact and flexible multisignature constructions. Additionally, the use of Merkleized Abstract Syntax Trees (MAST) allows multiple script conditions to be committed within a single transaction output, reducing on-chain footprint while preserving verifiability. Despite these capabilities, Taproot has not yet been leveraged to design a complete protocol for decentralized document verification that simultaneously supports integrity, issuer authenticity, and revocation.

4.1 Research Questions

This work aims to address the following research questions:

- How can Taproot be used to build a decentralized document verification protocol on Bitcoin?
- How does using Merkle tree commitments in Taproot improve privacy and scalability during document verification?
- How can document revocation be supported within a Taproot-based verification protocol?

4.2 Specific Objectives

To answer the proposed research questions, the following objectives are defined:

a) To design a Merkle tree structure that enables the verification of document integrity.

b) To design a mechanism for verifying issuer authenticity using digital signatures.

c) To design a revocation tracking mechanism that allows the verification of document validity.

d) To propose a proof of concept (PoC) of the proposed protocol on a Bitcoin test network.

e) To estimate the protocol performance in terms of scalability, computational complexity, and on-chain costs measured in virtual bytes (vBytes).

5 Methodology

To achieve the objectives, we propose a methodology divided into three phases: (1) protocol design and architectural specification, (2) proof-of-concept (PoC) prototyping, and (3) experimental validation and analysis.

5.1 Phase I: Protocol Design and Architecture

In this phase, objectives (a), (b), and (c) are addressed by specifying the protocol components, data structures, and verification logic. The protocol defines two roles:

- **Issuer**: the organization that issues a document and generates an associated *validation proof* delivered to the holder.
- **Verifier**: any third party that receives the document and its validation proof and performs verification autonomously.

Verification requires access only to: (i) the Bitcoin blockchain history, (ii) the document, and (iii) the validation proof. No interaction with the issuing organization is required at verification time.

5.1.1 On-chain vs Off-chain Components

The protocol is designed to minimize on-chain footprint while enabling self-contained verification.

- **On-chain**: a Bitcoin transaction anchors a batch commitment using a single pay-to-taproot (P2TR) output that commits to a Merkle root (MAST). Revocation updates are published, when needed, through spending transactions that encode a revocation state.

- **Off-chain**: document hashing, Merkle proof reconstruction, output key derivation (TapTweak), signature verification, and revocation state interpretation are performed by the verifier locally.

Figure 1 presents a high-level view of the protocol architecture, illustrating the interaction between issuer, holder, verifier, and the Bitcoin blockchain, as well as the separation between on-chain commitments and off-chain verification logic.

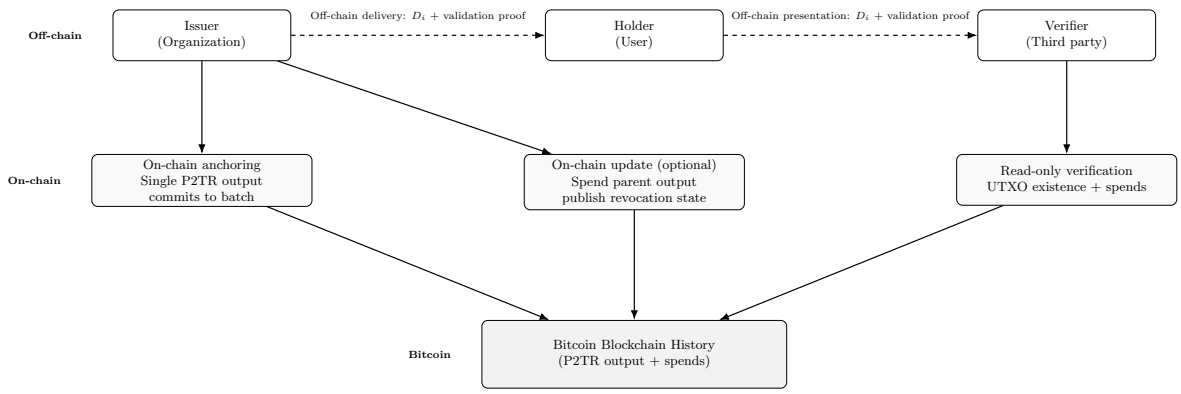

Figure 1: High-level architecture and data flow of the proposed Taproot-based document verification protocol. Dashed arrows represent off-chain data exchange, while solid arrows denote on-chain interactions.

The protocol considers a batch of size N, where each document D_i is associated with an index i (denoted D_i or $\mathsf{idx} = i$). A document is considered *valid* if and only if all three checks succeed:

- **Integrity verification**: the document content matches an on-chain commitment.

- **Issuer authenticity verification**: the issuer's signature is valid under a public key bound to the on-chain commitment.

- **Revocation verification**: the document index is not marked as revoked in the latest published revocation state.

5.1.2 Integrity Verification

To preserve privacy, integrity is verified using cryptographic commitments derived from SHA-256 hashes. For each issued document D_i, the issuer computes:

$$h_i = \mathrm{SHA256}(D_i).$$

The set of hashes $\{h_1, \ldots, h_N\}$ is arranged into a Merkle tree, with each document represented by a separate Script leaf embedding its corresponding hash. Let m denote the resulting Merkle root. The commitment is then embedded into a Taproot output using the BIP341 tweak procedure, producing an output key Q derived from the issuer's internal Schnorr public key P and the Merkle root m.

For completeness, we follow the BIP341 tagged-hash definition for the tweak:

$$t = \mathrm{SHA256}(\mathrm{SHA256}(\text{"}\texttt{TapTweak}\text{"}) \parallel \mathrm{SHA256}(\text{"}\texttt{TapTweak}\text{"}) \parallel P \parallel m), \tag{1}$$

and the tweaked output key is computed as:

$$Q = P + (t \cdot G),$$

where G is the generator point of secp256k1. The P2TR output on-chain commits to the entire batch through m, without revealing individual documents.

Leaf script design (to be analyzed). Each Merkle leaf corresponds to a script commitment that binds a single document hash h_i (and optionally its index $\mathsf{idx} = i$) within a *provably unspendable* script. The exact leaf construction will be evaluated in the thesis with respect to standardness, privacy, and implementation simplicity. A representative form is:

$$S_{leaf,i} = \texttt{OP_RETURN} \parallel D_{commit,i}, \quad \text{where} \quad D_{commit,i} = h_i \parallel i.$$

This design ensures that the script path does not enable spending; instead, it serves solely as an information commitment.

Validation proof contents (integrity). For a given document index i, the issuer provides a validation proof containing the minimal information required to reconstruct

the Merkle root and derive the committed Taproot output key, including: (i) the Merkle authentication path for leaf i, (ii) the leaf-level commitment preimage required by the verifier (e.g., h_i and i), and (iii) the issuer internal public key P (or sufficient information to derive it, depending on the instantiation).

Integrity check (verifier). The verifier computes $h_i = \text{SHA256}(D_i)$, reconstructs m using the Merkle path, derives Q through TapTweak, and checks that the corresponding P2TR output exists in the blockchain history and matches the anchored transaction output. This check is performed off-chain and reveals no information about other documents in the batch.

5.1.3 Key Separation

The protocol uses two distinct key pairs:

- $Keys_{btc}$ for controlling and spending Bitcoin funds (key-path spend of the P2TR output).
- $Keys_{sign}$ for signing documents off-chain.

This separation reduces exposure risks: Sk_{sign} may be managed by an institutional signing service, while Sk_{btc} can be protected independently (e.g., hardware-backed custody).

Each batch binds its own Pk_{sign} through its committed Merkle structure. Therefore, historical proofs remain verifiable under the historical signing key.

5.1.4 Issuer Authenticity Verification (Signature)

Issuer authenticity is verified using a Schnorr digital signature over the document hash (or an agreed message derived from it). To reduce metadata exposure, the issuer's signing public key is *not* published via `OP_RETURN`. Instead, the signing public key Pk_{sign} is *cryptographically bound* to the on-chain Taproot commitment.

Concretely, Pk_{sign} is included in the data committed by the leaf (or committed at the tree level), such that any modification of Pk_{sign} would change the Merkle root m and therefore change the derived output key Q. This makes Pk_{sign} verifiable by reconstruction, while remaining hidden on-chain unless the script path is revealed by spending.

Signature generation (issuer). For each document D_i, the issuer computes $h_i = \text{SHA256}(D_i)$ and generates a Schnorr signature:

$$\sigma_i = \text{Sign}(Sk_{sign}, h_i),$$

where Sk_{sign} is the issuer's signing private key. The signature σ_i is included in the validation proof delivered to the user.

Authenticity check (verifier). Given $(D_i, \sigma_i, Pk_{sign})$ from the validation proof, the verifier checks:

$$\text{Verify}(Pk_{sign}, h_i, \sigma_i) = \texttt{true}.$$

Additionally, the verifier confirms that Pk_{sign} is consistent with the on-chain commitment by reconstructing the Taproot output key Q from the proof. This achieves non-repudiation while avoiding explicit publication of Pk_{sign} via `OP_RETURN`.

5.1.5 Document Revocation

The protocol supports revocation through a state commitment published in a spending transaction. To revoke documents in a batch, the issuer spends the batch's P2TR output via the key path using Sk_{btc}, and publishes an updated revocation state.

A compact representation is a bitmap R of length N, where:

$$\mathsf{R}[i] = 1 \Rightarrow D_i \text{ revoked}, \qquad \mathsf{R}[i] = 0 \Rightarrow D_i \text{ valid}.$$

The updated bitmap is anchored in the child transaction (e.g., via `OP_RETURN` as a revocation record), and the verifier queries the latest spending transaction of the parent output to obtain the most recent revocation state.

Revocation is defined at the batch level: a single spending output updates the state of the entire batch. This design avoids requiring changes to the original Merkle commitments. To ensure unambiguous lookup, the protocol binds the document index i to the corresponding leaf commitment; alternative constructions and their impact on robustness and verification complexity are analyzed in the thesis.

Figure 2 summarizes the complete off-chain verification workflow executed by the verifier.

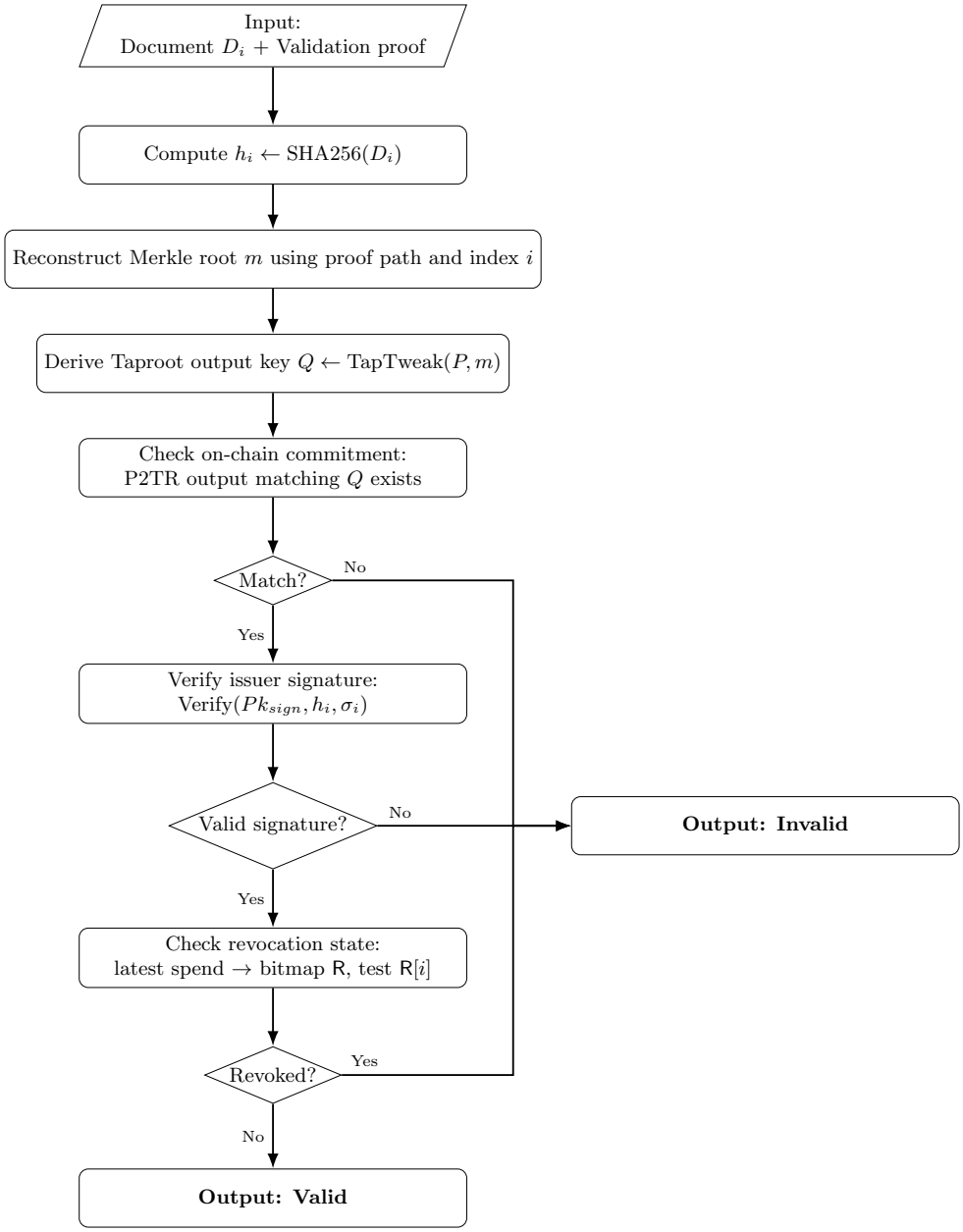

Figure 2: Off-chain verification procedure for a document and its validation proof, including integrity, authenticity, and revocation checks.

5.2 Phase II: Proof of Concept (PoC)

In this phase, objective (d) is addressed by proposing a PoC on Bitcoin test networks (Signet or Testnet), avoiding economic cost during development. The prototype will be designed in Python and will interface with Bitcoin Core running on Debian Linux. All

steps will be documented to support reproducibility and to allow any verifier to validate documents without third-party services.

The PoC will define concrete file formats for delivery:

- The issued document (e.g., `.pdf`).
- The validation proof (e.g., a structured `.json` file) containing Merkle path data, committed keys, signatures, and transaction references.

5.3 Phase III: Performance Estimation and Analysis

Performance estimations will be conducted using the PoC by varying the batch size N and the number of revocation updates. Objective (e) is addressed by measuring:

- **Scalability**: size of the validation proof and verification runtime as a function of N. Using Merkle trees, proof reconstruction is expected to be bounded by $O(\log N)$.
- **On-chain cost**: transaction cost measured in virtual bytes (vBytes), analyzed as a function of batch size and revocation frequency.

Finally, the results will be used to characterize the trade-offs between privacy, scalability, batch size, and network cost. The proposed protocol will be compared against legacy approaches, such as `OP_RETURN`-based methods, considering:

- Required on-chain bytes,
- Verification model (on-chain vs. off-chain),
- Privacy properties,
- Revocation support,
- Scalability behavior.

6 Timeline

The following timeline outlines the execution plan of the project described in Section 5. It is structured into four sequential and partially overlapping phases, covering a total duration of seventeen weeks.

- **Weeks 1–7**: Design and formal specification of the decentralized document verification protocol based on Taproot.
- **Weeks 6–12**: Proposal for a Proof of Concept (PoC) on a Bitcoin test network.
- **Weeks 13–14**: Performance and security evaluation of the proposed protocol.
- **Weeks 15–17**: Writing, revision, and preparation of the final thesis document and presentation.

The partial overlap between the first two phases reflects the iterative nature of protocol design and testing, allowing early prototyping to inform architectural decisions.

6.1 Gantt Chart Representation

Figure 3 presents the proposed schedule using a Gantt chart, providing a visual overview of the project phases and their respective durations.

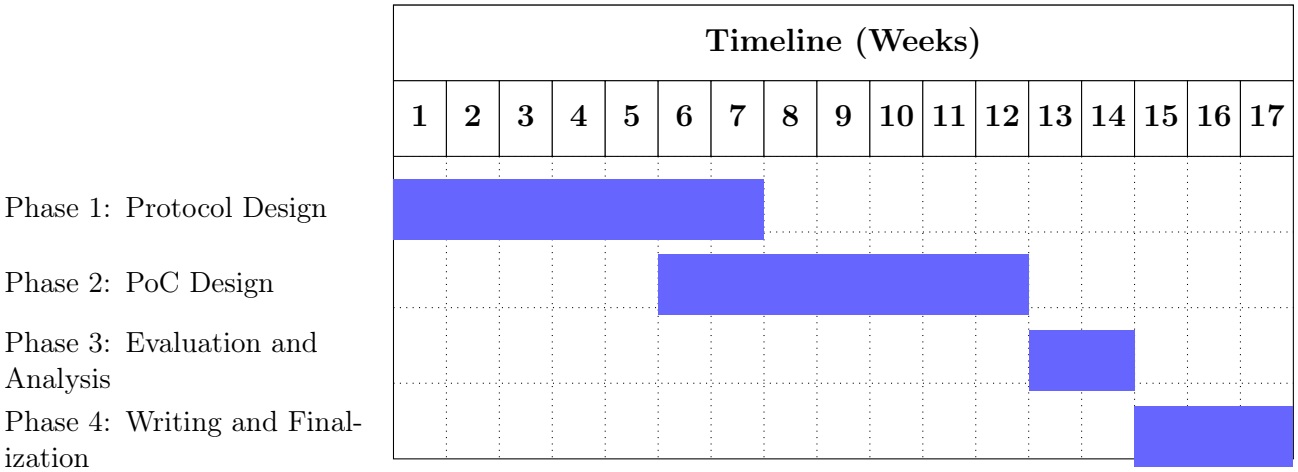

Figure 3: Gantt chart of the proposed project timeline

7 Conclusion

Decentralized document verification protocols offer clear advantages over traditional centralized systems. Properties such as high availability, resistance to denial-of-service attacks, and immutability are inherent to the Bitcoin network and make it a suitable foundation for long-term verification mechanisms. However, existing approaches studied in the literature reveal that Bitcoin-based solutions have historically faced limitations in

scalability, privacy, and expressiveness when applied to document verification. Protocols such as `OP_RETURN`-based anchoring and OpenTimestamps exhibit structural constraints that restrict their applicability to large-scale or privacy-sensitive use cases.

With the activation of the Taproot upgrade in 2021, Bitcoin introduced new cryptographic primitives that significantly expand its design space. Despite these advancements, this work identifies that Taproot has not yet been explored in the literature as the basis for a complete decentralized document verification protocol. This observation motivates the investigation proposed in this thesis.

The protocol presented in this work leverages Taproot's cryptographic capabilities to support document integrity verification through Merkle tree commitments and issuer authenticity verification through Schnorr digital signatures. It further enables document revocation through transaction chaining and supports signing key rotation without invalidating previously issued documents. By anchoring documents in batches and avoiding explicit disclosure of auxiliary data, the protocol preserves privacy by preventing information leakage about other documents within the same batch.

This work explicitly limits its scope to cryptographic verification and does not address the semantic validity of document contents, which remains the responsibility of the issuing organization. Additionally, long-term preservation of documents and validation proofs after issuance is not considered. Future work includes the study of archival mechanisms for long-term proof persistence, enhanced revocation strategies, and incentive or penalty schemes for entities that attempt to present invalid or revoked documents as valid.

References

[1] Eric Lombrozo, Johnson Lau, and Pieter Wuille. *BIP 141: Segregated Witness (Consensus layer)*. Bitcoin Improvement Proposals. Accessed: 2025-11-20. 2015. URL: https://github.com/bitcoin/bips/blob/master/bip-0141.mediawiki.

[2] Pieter Wuille, Jonas Nick, and Anthony Towns. *BIP 341: Taproot: SegWit version 1 spending rules*. Bitcoin Improvement Proposals. Accessed: 2025-11-20. 2020. URL: https://github.com/bitcoin/bips/blob/master/bip-0341.mediawiki.

[3] Victoria Louise Lemieux. "Trusting records: is Blockchain technology the answer?" In: *Records Management Journal* 26.2 (2016), pp. 110–139. DOI: 10.1108/RMJ-12-2015-0042.

[4] Yuefei Gao and Hajime Nobuhara. "A Decentralized Trusted Timestamping Based on Blockchains". In: *IEEJ Journal of Industry Applications* 6.4 (2017), pp. 252–257. DOI: 10.1541/ieejjia.6.252.

[5] Andrew Sward, Ivy Vecna, and Forrest Stonedahl. "Data Insertion in Bitcoin's Blockchain". In: *LEDGER* 3 (2018), pp. 1–23. DOI: 10.5915/LEDGER.2018.101.

[6] Massimo Bartoletti, Bryn Bellomy, and Livio Pompianu. "A Journey into Bitcoin Metadata". In: *J Grid Comput* 17 (2019), pp. 3–22. DOI: 10.1007/s10723-019-09473-3.

[7] Elias Strehle and Fred Steinmetz. "Dominating OP Returns: The Impact of Omni and Veriblock on Bitcoin". In: *J Grid Comput* 18 (2020), pp. 575–590. DOI: 10.1007/s10723-020-09537-9.

[8] Blockcerts Project. *Blockcerts: Open Standard for Blockchain Credentials*. Especificación del estándar desarrollado originalmente por el MIT Media Lab. 2024. URL: https://www.blockcerts.org/ (visited on 11/27/2025).

[9] Todd, Peter. *OpenTimestamps*. Herramienta para probar la existencia de archivos usando pruebas de Merkle en Bitcoin. 2024. URL: https://opentimestamps.org/ (visited on 11/27/2025).

[10] Michal Kedziora et al. "Analysis of segregated witness implementation for increasing efficiency and security of the Bitcoin cryptocurrency". In: *Journal of Information and Telecommunication* 7.1 (2023). Published online: 26 Sep 2022, pp. 44–55. DOI: 10.1080/24751839.2022.2122301.

[11] Víctor Gayoso Martínez, Luis Hernández-Álvarez, and Luis Hernández Encinas. "Analysis of the Cryptographic Tools for Blockchain and Bitcoin". In: *Mathematics* 8.1 (2020), p. 131. DOI: 10.3390/math8010131.

[12] Weidong Fang et al. "Digital signature scheme for information non-repudiation in blockchain: a state of the art review". In: *EURASIP Journal on Wireless Communications and Networking* 2020.56 (2020), pp. 1–15. DOI: 10.1186/s13638-020-01665-w.

[13] Xiao Yang et al. "Efficient Verifiably Encrypted ECDSA-Like Signatures and Their Applications". In: *IEEE Transactions on Information Forensics and Security* 17 (2022), pp. 2001–2016. DOI: 10.1109/TIFS.2022.3165978.

[14] Peng Zhang et al. "Achieving High Efficiency in Schnorr-Based Multi-Signature Applications in Blockchain". In: *Electronics* 14.9 (2025), p. 1883. DOI: 10.3390/electronics14091883.

[15] Pieter Wuille, Jonas Nick, and Tim Ruffing. *BIP 340: Schnorr Signatures for secp256k1*. Bitcoin Improvement Proposals. Accessed: 2025-11-20. 2020. URL: https://github.com/bitcoin/bips/blob/master/bip-0340.mediawiki.

[16] Ze Wang et al. "Blockchain-Based Certificate Transparency and Revocation Transparency". In: *Financial Cryptography and Data Security: FC 2018 International Workshops, BITCOIN, VOTING, and WTSC, Revised Selected Papers*. Ed. by Aviv Zohar et al. Vol. 10958. Lecture Notes in Computer Science. Springer, 2019, pp. 144–164.

[17] Yannan Li et al. "Decentralized Public Key Infrastructures atop Blockchain". In: *IEEE Network* (2020). DOI: 10.1109/MNET.011.2000085.

[18] Diego Pennino et al. "Efficient Certification of Endpoint Control on Blockchain". In: *IEEE Access* 9 (2021), pp. 131901–131915. DOI: 10.1109/ACCESS.2021.3115343.

[19] Tonino Palmisano et al. "Notarization and Anti-Plagiarism: A New Blockchain Approach". In: *Appl. Sci.* 12.1 (2022), p. 243. DOI: 10.3390/app12010243.

[20] Ayush Mishra et al. "Blockchain-Based Decentralized Document Verification and Its Applications". In: *Journal of Information Systems Engineering and Management* 10 (10s 2025). ISSN: 2468-4376.

[21] Felix Härer and Hans-Georg Fill. "Decentralized Attestation and Distribution of Information Using Blockchains and Multi-Protocol Storage". In: *IEEE Access* 10 (2022), pp. 20017–20032. DOI: 10.1109/ACCESS.2022.3150356.

[22] Diana Hawashin et al. "Blockchain and NFT-based traceability and certification for UAV parts in manufacturing". In: *Journal of Industrial Information Integration* 39 (2024), p. 100597. DOI: 10.1016/j.jii.2024.100597.

[23] Renato Q. Castro and Manuel Au-Yong-Oliveira. "Blockchain and Higher Education Diplomas". In: *Eur. J. Investig. Health Psychol. Educ.* 11.1 (2021), pp. 154–167. DOI: 10.3390/ejihpe11010013.

[24] Avni Rustemi et al. "A Systematic Literature Review on Blockchain-Based Systems for Academic Certificate Verification". In: *IEEE Access* 11 (2023), pp. 66498–66518. DOI: 10.1109/ACCESS.2023.3289598.

[25] Šeila Bećirović Ramić et al. "Selective disclosure in digital credentials: A review". In: *ICT Express* 10 (2024), pp. 916–934. DOI: 10.1016/j.icte.2024.05.011.

[26] Tiantian Zhang et al. "Privacy Protection During the Issuance and Revocation of Verifiable Credentials in Self-Sovereign Identity". In: *Concurrency and Computation: Practice and Experience* 37.9 (2025), e70084. DOI: 10.1002/cpe.70084.